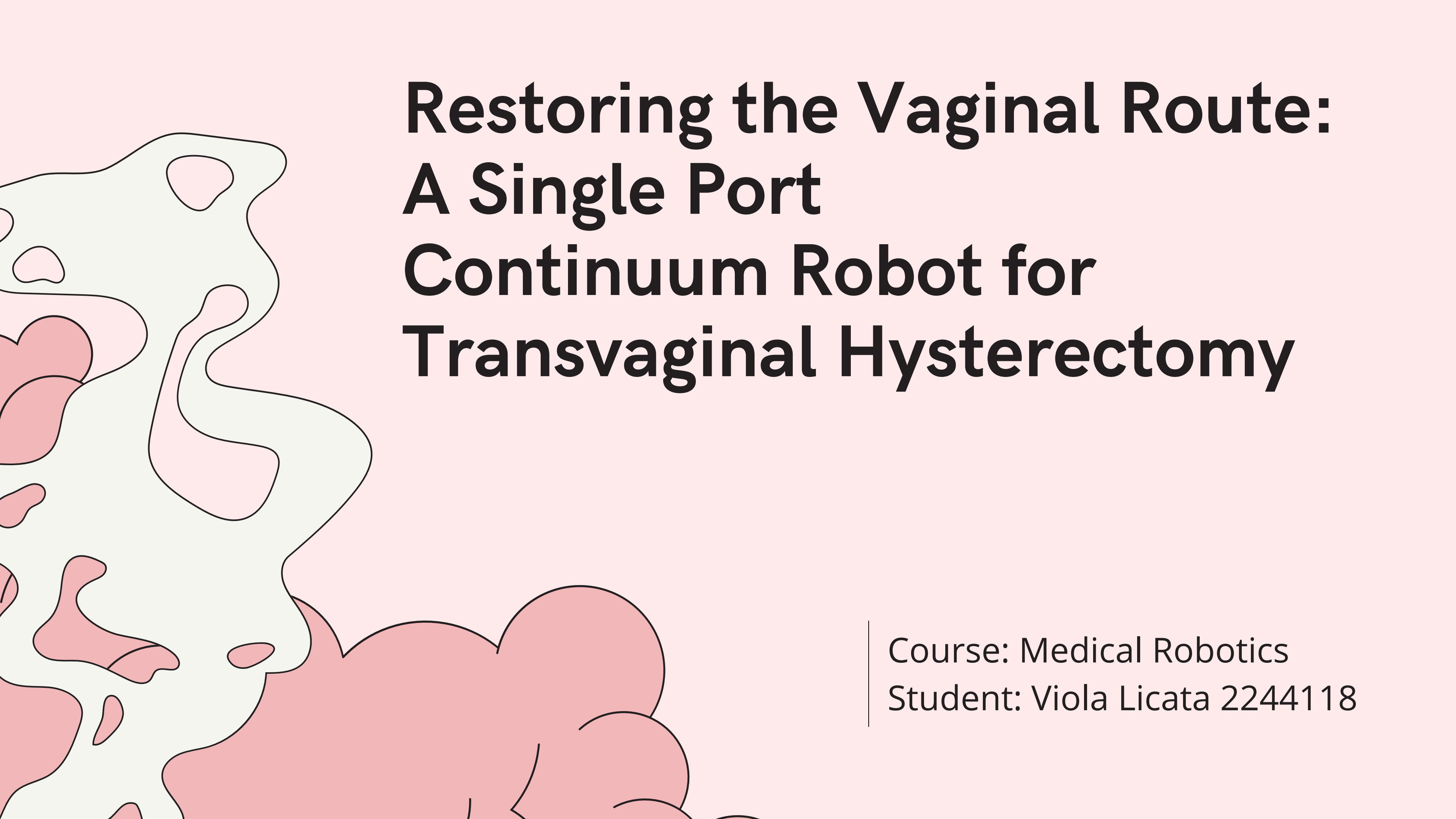




Restoring the Vaginal Route: A Single Port Continuum Robot for Transvaginal Hysterectomy

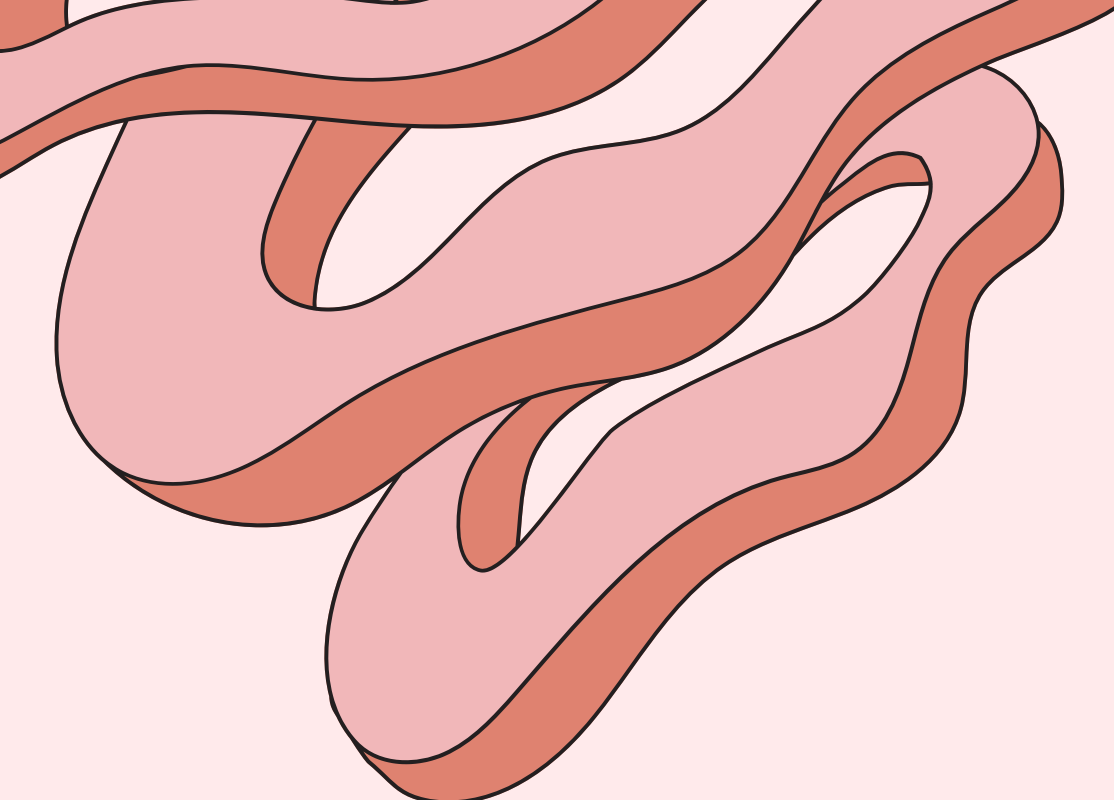
Course: Medical Robotics
Student: Viola Licata 2244118

The Problem:

Over 800,000 women each year undergo invasive abdominal surgery for procedures that could be performed through a natural orifice.

The Need:

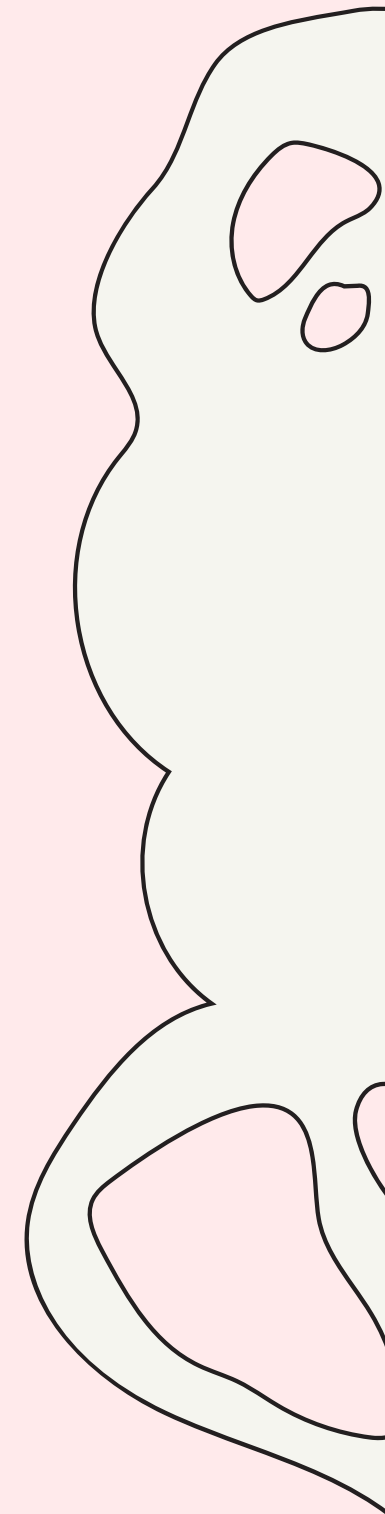
The lack of suitable instruments forces surgeons into abdominal approaches, increasing patient risk, surgical trauma, and healthcare costs. The goal is to make natural-orifice hysterectomy feasible.

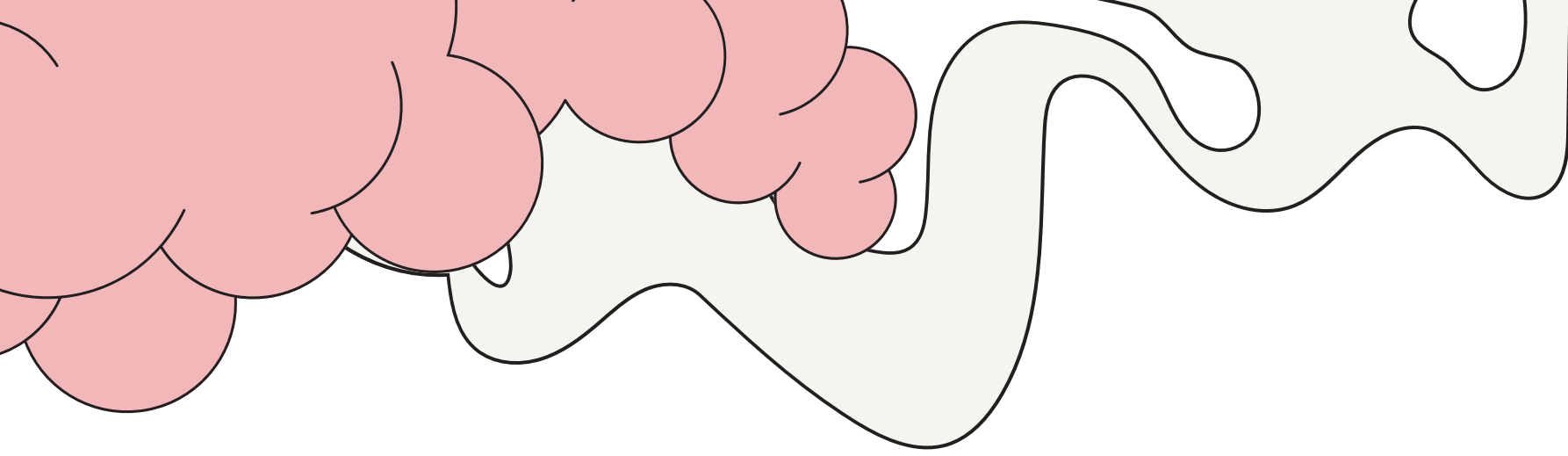


Why It Remains Unsolved?

The vaginal route is a narrow, conical channel that forces standard rigid tools into parallel alignment, causing constant tool clashing and surgeon fatigue.

Current robots fail in this space: multi-port systems cause abdominal wall trauma, while single-port rigid platforms require straight clearance that causes severe collisions in the pelvic cavity.



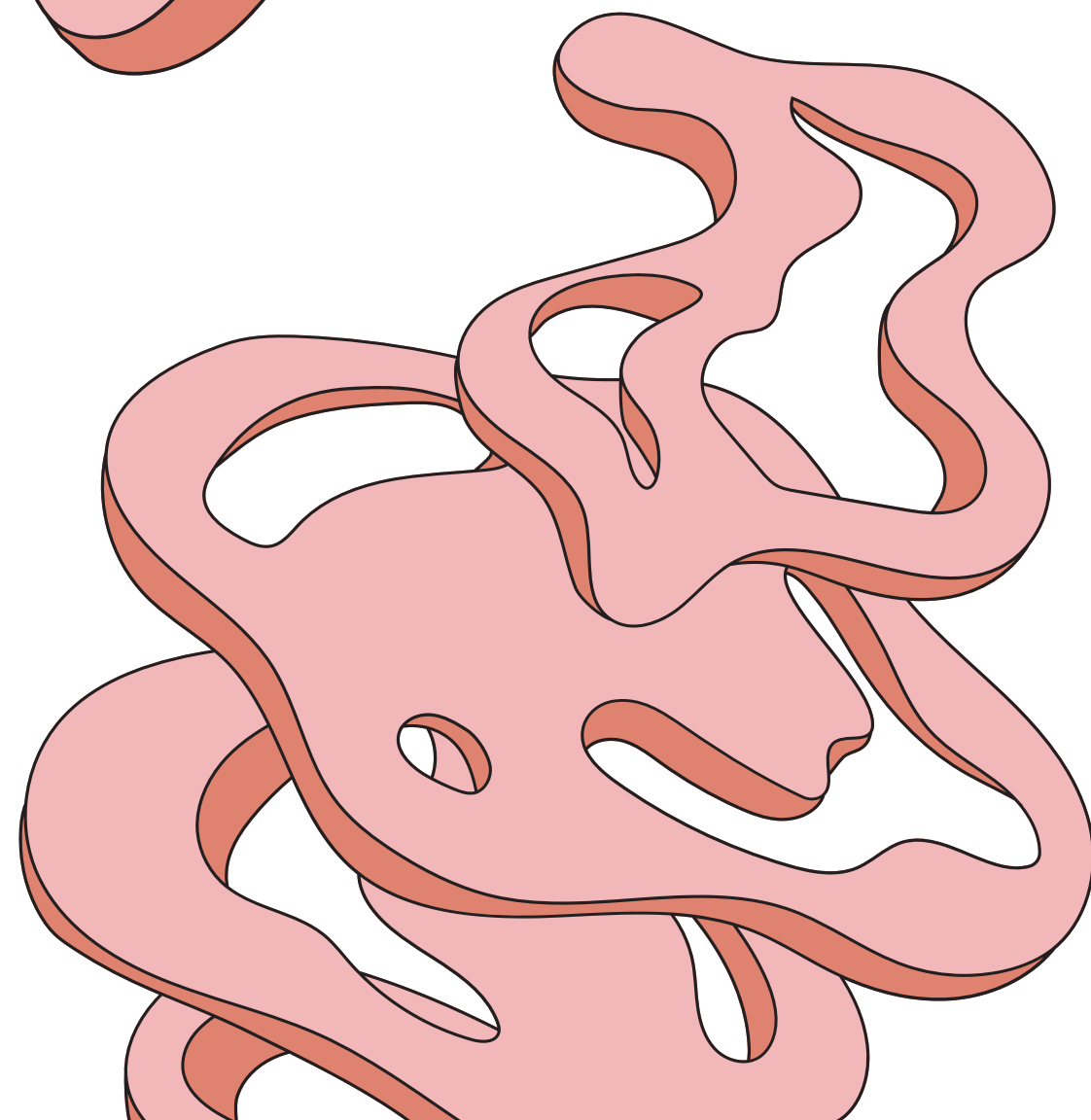
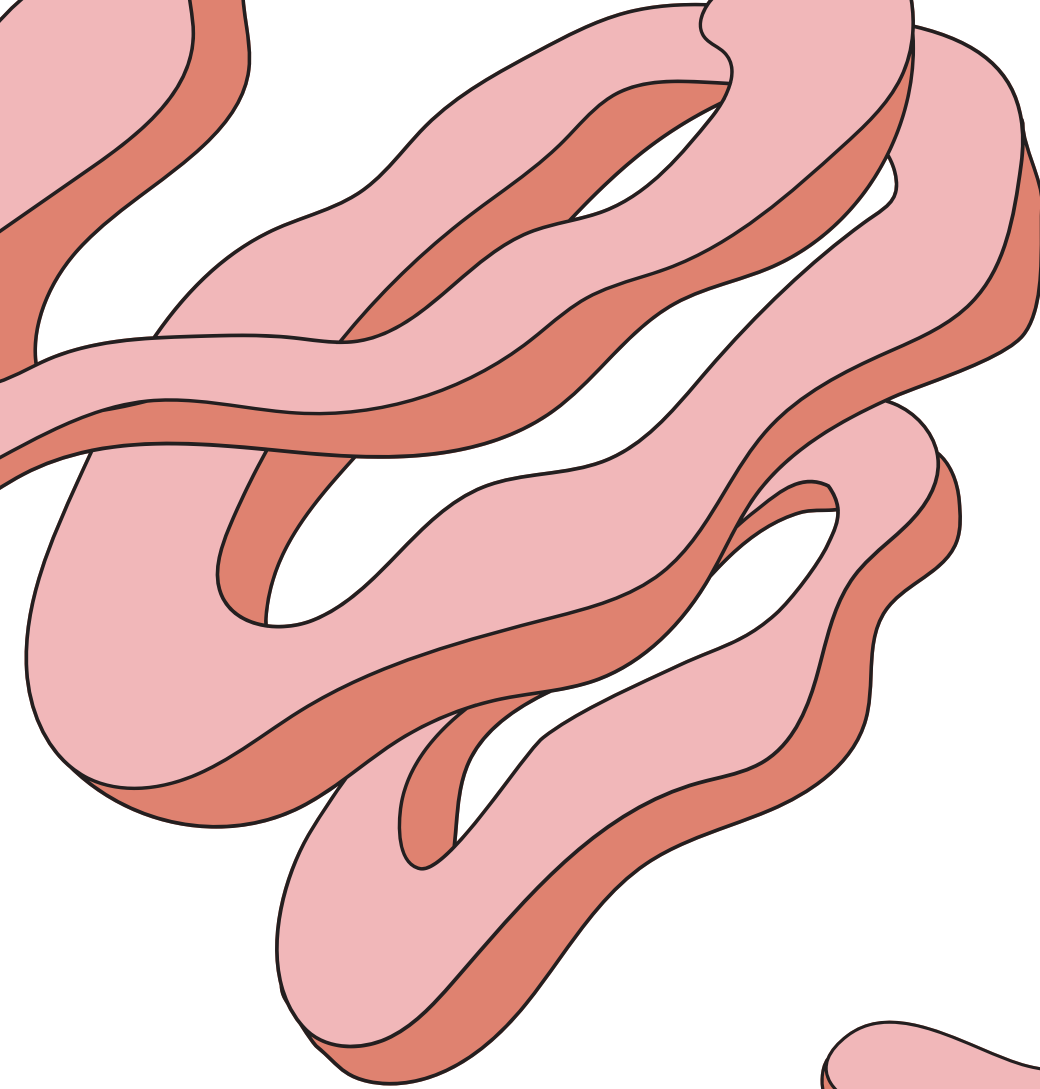


Single-Port Continuum Platform: An 18 mm flexible overtube delivering two tendon-driven Nitinol continuum arms and a 3D stereoscopic camera directly into the pelvic cavity.

Proposed Robotic Solution

Active Triangulation & Safety: The continuum arms expand outward in an S-shape to recover a 50 degree internal triangulation angle, while integrated FBG optical sensors detect abnormal tissue contact to prevent perforation





Thank you!

